

DUNDRENNAN, United Kingdom

WMO: 031530

Lat: 54.8033N

Lon: 4.0081W

Elev: 371

StdP: 14.50

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 27.7	(c) 29.9	(d) 21.7	(e) 16.5	(f) 32.0	(g) 23.9	(h) 18.3	(i) 33.3	(j) 44.1	(k) 43.7	(l) 40.3	(m) 43.5	(n) 9.6	(o) 20	(p) 0.761

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.2	(c) 70.2	(d) 61.9	(e) 67.0	(f) 60.3	(g) 64.6	(h) 58.9	(i) 63.5	(j) 67.9	(k) 61.7	(l) 65.1	(m) 60.4	(n) 63.0	(o) 10.0	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.7	83.3	64.9	60.4	79.4	62.8	59.3	76.3	61.5	29.0	67.9	27.8	65.2	26.8	63.2	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature										
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min		Max		Min	Max	Min		Max	Min		Max	Min		Max	Min		Max
(a)	(b)	(c)	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
34.9	31.2	28.0 <td>WB</td> <td>24.7</td> <td>76.0</td> <td>2.9</td> <td>3.8</td> <td>22.7</td> <td>78.8</td> <td>21.0</td> <td>81.0</td> <td>19.4</td> <td>83.1</td> <td>17.3</td> <td>85.9</td>		WB	24.7	76.0	2.9	3.8	22.7	78.8	21.0	81.0	19.4	83.1	17.3	85.9	
				24.2	67.5	2.6	2.6	22.4	69.4	20.8	70.9	19.4	72.4	17.5	74.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	39.8	40.3	41.8	45.3	50.4	54.9	58.0	58.0	55.1	50.3	44.7	40.8
	DBStd	7.96	4.62	4.13	4.20	4.13	4.53	3.92	3.53	2.86	3.44	4.40	4.97	5.48
	HDD50	1547	315	273	255	147	47	3	0	0	4	51	166	286
	HDD65	6093	780	693	720	590	453	303	220	218	299	457	609	751
	CDD50	938	0	0	1	7	59	152	249	248	156	59	7	0
	CDD65	6	0	0	0	0	0	1	4	1	0	0	0	0
	CDH74	28	0	0	0	0	0	7	20	1	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	13.8	16.2	15.7	14.7	13.2	13.1	11.7	10.9	11.6	12.9	14.8	14.9	15.3
Precipitation	PrecAvg	54.1	5.6	4.0	4.1	3.1	3.1	3.2	3.5	4.3	4.3	6.1	5.7	6.1
	PrecMax	65.9	9.8	7.9	7.0	5.6	5.6	8.0	5.7	10.2	8.1	10.3	11.0	13.1
	PrecMin	41.2	1.2	0.7	1.9	1.2	0.6	1.2	1.4	0.7	1.1	1.4	2.7	1.7
	PrecStd	7.6	2.2	2.1	1.2	1.1	1.5	1.6	1.2	2.1	1.7	2.8	2.2	2.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	49.9	55.9	62.5	70.3	73.7	77.1	71.1	67.2	60.4	55.8	52.3
		MCWB	49.0	47.2	49.1	52.5	60.5	62.3	66.2	62.9	61.1	56.9	54.5	50.9
	2%	DB	48.6	48.4	51.2	56.7	65.1	68.2	71.1	67.4	63.8	58.7	54.2	50.7
		MCWB	47.3	46.5	46.9	50.0	57.5	60.2	62.7	61.3	58.5	56.6	52.9	49.7
	5%	DB	47.5	47.1	48.9	53.5	61.2	64.4	67.2	64.9	61.7	57.5	52.9	49.6
		MCWB	46.3	45.5	46.0	48.3	54.7	57.6	60.9	60.0	57.9	55.7	51.6	48.5
	10%	DB	46.4	46.0	47.4	51.2	57.7	61.7	64.2	62.9	60.2	56.4	51.7	48.3
		MCWB	45.2	44.5	45.2	47.4	52.3	56.4	59.2	58.8	57.2	54.6	50.4	47.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	48.5	51.0	54.1	61.9	64.2	67.3	64.8	62.7	58.9	55.3	51.5
		MCDB	50.1	48.9	53.7	60.7	68.3	70.4	74.7	68.6	65.2	59.4	55.6	51.9
	2%	WB	48.0	47.1	48.3	51.2	58.6	61.1	64.1	62.7	60.4	57.5	53.4	50.1
		MCDB	48.5	47.8	49.8	54.7	63.7	66.7	69.5	65.7	61.9	58.1	53.9	50.5
	5%	WB	46.7	46.0	46.8	49.5	55.7	59.0	61.9	61.2	59.2	56.3	52.0	48.8
		MCDB	47.3	46.8	48.0	52.0	59.8	63.2	65.6	63.3	60.7	57.1	52.6	49.5
	10%	WB	45.4	44.9	45.7	48.2	53.4	57.2	60.2	60.1	58.0	55.0	50.6	47.3
		MCDB	46.2	45.7	47.0	50.4	56.5	60.5	63.3	61.9	59.6	56.1	51.5	48.2
Mean Daily Temperature Range	5% DB	MDBR	6.3	7.0	8.5	10.0	11.0	10.3	10.2	9.4	8.9	7.9	7.2	6.8
		MCDBR	6.8	7.1	9.8	13.1	15.8	14.8	15.1	12.3	10.4	7.5	7.1	7.3
		MCWBR	7.2	6.6	7.4	8.7	10.1	9.3	9.3	8.0	7.1	6.9	7.7	7.9
	5% WB	MCDBR	6.7	6.4	8.4	10.6	13.5	12.7	12.6	9.6	8.3	7.0	6.7	7.3
		MCWBR	7.3	6.6	7.3	7.9	9.5	8.7	8.8	7.3	6.9	7.3	7.6	8.0
Clear-Sky Solar Irradiance	taub		0.325	0.336	0.360	0.378	0.381	0.377	0.378	0.365	0.356	0.341	0.318	0.310
	taud		2.272	2.364	2.337	2.317	2.358	2.395	2.394	2.444	2.472	2.452	2.352	2.248
	Ebn at Noon		189	236	257	268	274	276	273	270	258	232	193	166
	Edh at Noon		19	24	30	36	36	35	35	31	27	22	17	15
All-Sky Solar Radiation	RadAvg		173	386	747	1202	1551	1551	1491	1214	852	462	223	129
	RadStd		22	41	64	105	127	153	144	90	65	54	23	13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page